DPUSGETextureDirectShow

Creates a dynamic video texture in the virtual world. The texture contains a live video image that is captured from a DirectShowcompatible camera, such as a webcam.

1. Dynamic parameters:

VFlip	Flips the video vertically, if the value is set to 1. The default value 0 doesn't flip the video.
Start	Starts video update when the value changes from 0 to 1. Default: 1 (starts video update immediately when the simulation is started)
Stop	Stops video update when the value changes from 0 to 1.

2. Static parameters:

View	Name of the SGE view for which the video texture is updated. Additionally, it will be visible in all views that share textures with the specified SGE view. Default: *SGEView*
TextureName	Name of the created video texture, which must start with a dollar sign (*$*). Default: *$Video*
VideoSource	Can be used to select an external video source, if more than one is connected to the PC. Specify the name of the desired video source (or an unambiguous part of it). The names of all connected video sources are displayed if *VideoSource=XXX* is specified.
VideoSourceNumber	Selects one of the connected video sources with the same name (if available). Default: 1 (selects the first source having the *VideoSource* name)
VideoW, *VideoH*	Desired width and height of the video captured from the camera. If not specified, a camera-dependent default value is used. **Warning**: Most cameras accept only a number of fixed video formats. A similar format will be selected if the desired format is not available. Additionally, some formats may not work correctly although they are advised by the camera's driver.
VideoFPS	Desired frame rate of the video captured from the camera. If not specified, a camera-dependent default value is used. **Warning**: Most cameras accept only a number frame rate. A similar frame rate will be selected if the desired one is not available. Additionally, some frame rates may not work correctly although they are advised by the camera's driver.

Default: 0 (stops video update only when the simulation is stopped)

3. Functional principle:

Using *DPUSGETextureDirectShow*, a dynamic video texture is created in the virtual world. The texture contains a live video image that is captured from a DirectShow compatible camera. (Typical USB webcams should work.)

The *TextureName* parameter defines a name under which the video texture will be available in the SGE world. The name must start with a dollar sign ($). The video texture can be used in any SGE object loaded after this DPU by using this texture name instead of specifying a texture file on the hard disk. Alternatively, *DPUSGEVideoObject* can be used to create a simple overlay object which displays the video texture.

4. Example of a SGE object which references a video texture:

```
define SIG3DObject VideoObj
{
      AlphaTest = 0.2;
      Blend = false;
      // [...]
      Texture = $MyVideo;
      TextureFilter = NoCompression | ClampS | ClampT | MagLinear |
                        MinLinear | Tex-ture;
      // [...]
};
```

5. Dependencies to other DPUs:

Please make sure that the computer on which **DPUSGETextureDirectShow** runs also executes **DPUSGEWorld**, **DPUSGEObjectXXX** and **DPUSGEView**. The Index values of these DPUs must be set in the following order:

DPUSGEWorld.Index < DPUSGETextureDirectShow.Index < DPUSGEOb-jectXXX.Index < DPUSGEView.Index

6. Example:

This DPU works well using the following script:

```
DPUSGETextureDirectShow VidTexture
{
      Computer = {VIS};
      Index = 50;

      TextureName = "$MyVideo";
      # Use Bayer pixel encoding for optimum transfer speed
      PixelFormat = "BayerRG8";
```

```
};
```